

Identifying Data Requirements

Project Title: Online Shopping Database Management System (Implementation: SmartShop)

1. Customer Entity

Description

The Customer entity stores information about registered users who purchase products from the online shopping platform.

ATTRIBUTE	DESCRIPTION
CUSTOMERID (PK)	Unique customer identifier
NAME	Customer's full name
EMAIL	Customer's email address
PASSWORD	Encrypted login password
PHONENUMBER	Contact number
ADDRESS	Delivery address
REGISTRATIONDATE	Date of account creation

Primary Key: CustomerID

2. Product Entity

Description

The Product entity stores information about products available for sale.

ATTRIBUTE	DESCRIPTION
PRODUCTID (PK)	Unique product identifier
PRODUCTNAME	Name of the product
DESCRIPTION	Product details
PRICE	Product price
STOCKQUANTITY	Available stock
CATEGORYID (FK)	Product category
SUPPLIERID (FK)	Product supplier

Primary Key: ProductID

3. Category Entity

Description

The Category entity groups products into different categories.

ATTRIBUTE	DESCRIPTION
CATEGORYID (PK)	Unique category identifier
CATEGORYNAME	Category name
DESCRIPTION	Category description

Primary Key: CategoryID

4. Supplier Entity

Description

The Supplier entity stores supplier information.

ATTRIBUTE	DESCRIPTION
SUPPLIERID (PK)	Unique supplier identifier
SUPPLIERNAME	Supplier name
CONTACTNUMBER	Contact number
EMAIL	Supplier email
ADDRESS	Supplier address

Primary Key: SupplierID

5. Order Entity

Description

The Order entity stores customer order information.

ATTRIBUTE	DESCRIPTION
ORDERID (PK)	Unique order identifier
CUSTOMERID (FK)	Customer who placed the order
ORDERDATE	Date of order
TOTALAMOUNT	Total order value
ORDERSTATUS	Pending, Confirmed, Shipped, Delivered

Primary Key: OrderID

6. Payment Entity

Description

The Payment entity stores payment details.

ATTRIBUTE	DESCRIPTION
PAYMENTID (PK)	Unique payment identifier
ORDERID (FK)	Related order
PAYMENTMETHOD	UPI, Card, Net Banking
PAYMENTDATE	Payment date
AMOUNT	Payment amount
PAYMENTSTATUS	Successful or Failed

Primary Key: PaymentID

7. Shipment Entity

Description

The Shipment entity stores delivery information.

ATTRIBUTE	DESCRIPTION
SHIPMENTID (PK)	Unique shipment identifier
ORDERID (FK)	Related order
TRACKINGID	Courier tracking number
COURIERNAME	Delivery company
SHIPMENTSTATUS	Shipped, In Transit, Delivered
DELIVERYDATE	Delivery date

Primary Key: ShipmentID

8. Review Entity

Description

The Review entity stores customer feedback for purchased products.

ATTRIBUTE	DESCRIPTION
REVIEWID (PK)	Unique review identifier
CUSTOMERID (FK)	Customer who reviewed
PRODUCTID (FK)	Reviewed product
RATING	Rating (1–5)
REVIEWTEXT	Customer comments
REVIEWDATE	Review submission date

Primary Key: ReviewID

Summary of Entities

ENTITY	PRIMARY KEY
CUSTOMER	CustomerID
PRODUCT	ProductID
CATEGORY	CategoryID
SUPPLIER	SupplierID
ORDER	OrderID
PAYMENT	PaymentID
SHIPMENT	ShipmentID
REVIEW	ReviewID
